Chapter 3
Fundamentals of Metal-Oxide Resistive Random Access Memory (RRAM)

David C. Gilmer and Gennadi Bersuker

Abstract Detailed operational and switching characteristics for metal-oxide resistive random access memory (RRAM) are presented, along with materials/vacancy engineering ramifications for the switching operations. The reported experimental data is consistent with a physical picture of the RRAM switching as caused by oxidation/reduction processes in a conductive filament formed through the metal-oxide material. The oxygen interstitial ion and vacancy profile resulting from the filament formation process establishes the initial structural properties controlling subsequent reset/set resistance switching operations. The microscopic description of these processes links the device electrical and material characteristics allowing for optimized material compositional profiles to improve device performance characteristics.

3.1 Introduction

Continued progress in currently used memory systems (high-density memory cell arrays, dynamic random access memories (DRAM), NAND, storage class memory (SCM), advanced embedded type applications, etc.) depend on continued advances toward low operation currents and voltages while increasing density and speed. In addition, the rising requirements for reducing power consumption in mobile applications with convenience of use have increased the efforts and focus toward technology development of new nonvolatile memories [1]. In this respect, the resistance switching random access memory (RRAM) technology presents an attractive option due to demonstrated potential for low-complexity/high-density/low-cost nonvolatile

D. C. Gilmer (✉)
Nantero, Inc., Austin, TX, USA
e-mail: david.gilmer@nantero.com

G. Bersuker
Aerospace Corporation, El Segundo, CA, USA

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memory cells with high-speed/low-energy operations and prospective ability to satisfy the requirements of various incumbent advanced scaled memory systems currently in use [2, 3]. A common characteristic within the large family of the metal-oxide-based resistance switching memory schemes is that their operating mechanisms involve either rearranging the atomic structure of the dielectric material (rendering it conductive) or movement of atoms in the dielectric (resulting in the formation of a conductive path), as opposed to the current incumbent memory technologies based on the electron movement and storage. This overview is focused on the transition metal-oxide filament-based type of RRAM (with emphasis on hafnium oxides [HfOx]), which offers promising opportunities for a variety of nonvolatile memory applications.

Metal-oxide-based filament-type RRAM utilizes the repeatable formation and rupture of a localized conductive filament through the metal oxide and has the unique attribute of area-independent resistance. This filament-based mechanism suggests an ultimate scaling advantage that is only limited to the active filament size, which potentially can be only a few nm. The mechanisms of filament-type resistive switching depend on the materials (dielectric and metal electrodes) employed in the fabrication of the memory cell and can involve more than one type of conduction mode. Filament-based metal-oxide RRAM schemes have been implemented with a variety of transition metal oxides (such as HfO_2, ZrO_2, Ta_2O_5, TiO_2) which have recently received considerable attention due to their demonstrated nanosecond, low-power (<pJ) switching with high (~trillion cycle) endurance and retention of >10 years at 200C using fabrication-friendly binary oxide dielectrics and metal electrodes [4–12]. Among the metal-oxide dielectrics in this group, HfO_2 has recently become mainstream in advanced transistor gate stack applications and has also become one of the stronger candidates for RRAM [4, 6, 9, 10, 13, 14]. One of the advantages in using HfO_2 for metal-oxide filament-based RRAM is the extensive research performed during ~1997–2007 in developing gate stacks toward manufacturability and productization of advanced logic using metal gate and HfO_2 as the "high-K" gate insulator, which enabled the detailed understanding of HfO_2 resistivity, conductivity, defects, and breakdown mechanisms [15–18]. In addition, HfO_2 has one of the stronger oxygen affinities among the transition metals, and thus, thermodynamically speaking, it is relatively stable and compatible with most of the commonly used "fab-friendly" metallic electrodes such as TiN, TaN, W, etc. [19–24].

This review focuses primarily on the HfO_2-based RRAM system, discussing its operation and optimization of device characteristics through materials (and defect/vacancy) engineering along with a discussion of the possible mechanisms involved in the observed repeatable resistance switching. Although some of the processes contributing to resistive switching in the HfO_2-based RRAM may be active in other metal-oxide RRAM systems, due to the material-specific nature of RRAM characteristics (such as relative oxygen affinities, valance states, and atom diffusivities), any conclusions or comparisons to other metal-oxide material systems must be made with caution.

3.2 Operational Characteristics for HfO_2-Based RRAM

Typically, RRAM cells employ a metal-insulator-metal (MIM) structure, which is often referred to as a 1-resistor (or 1R) configuration. The resistance change behavior in bipolar operation for filament-based RRAM switching, as observed for HfO_2, is shown in Fig. 3.1a where an ohmic low resistance “on” state (LRS) or a non-ohmic high resistance “off” state (HRS) can be related to the formed conductive filament (CF) or its rupture, respectively (Fig. 3.1b). To describe the material changes in the dielectric associated with this type of resistive switching, one should start with the CF formation, where a microscopic description of the CF features and metal-oxide defect/vacancy engineering that enable memory operations have recently been proposed [25, 26]. The CF forming process in HfO_2 can be discussed in terms of dielectric breakdown [4], where there is an abrupt formation of a localized region between the electrodes, within which the dielectric composition becomes more oxygen-deficient (or metal-rich), rendering this region conductive (the CF formation). The associated abrupt conductance change for this CF formation is observed by applying a voltage across the RRAM dielectric in the fresh (“virgin”) state, Fig. 3.2. The typical bipolar *dc-iv* switching operation are noted on the graph: (1) forming [the initial dielectric breakdown event leading to a CF with the voltage of the abrupt breakdown (forming) termed V_F]; (2) RESET operation [at opposite voltage polarity (than forming or set) for the bipolar operation [*as opposed to forming/reset/set all at same polarity in unipolar switching*]], where the device changes from the LRS to the HRS state due to rupture of the CF; and (3) SET operation [at opposite voltage polarity vs reset and similar polarity as forming], where the device changes from the HRS to the LRS state due to reformation of a continuous CF between the two electrodes.

The typical *dc-iv* switching operation parameters and device response in bipolar-operated RRAM devices are shown in Fig. 3.3. Similar well-behaved devices can repeatedly be switched between the LRS and HRS states for billions of cycles, with the corresponding switching operations being reset (LRS→HRS) and set (HRS→LRS) [6, 7, 9, 14, 26].

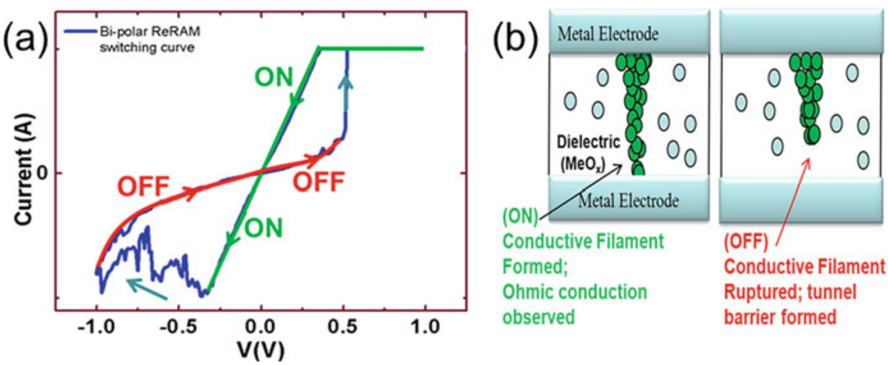

Fig. 3.1 (**a**) Characteristic dc properties of the on (LRS) and off (HRS) states and (**b**) related physical picture for filament-based RRAM

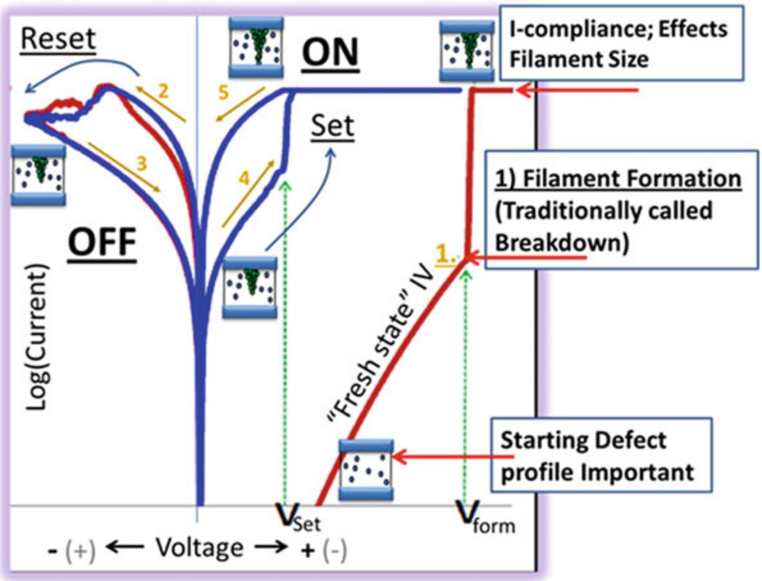

Fig. 3.2 Characteristic dc operation of HfO_2-based RRAM devices highlighting (1) forming [a dielectric breakdown event leading to a conductive filament (CF)]; (2) RESET operation, where the device changes from the LRS to the HRS state due to rupture of the CF; and (3) SET operation, where the device changes from the HRS to the LRS state due to reformation of the CF path

3.2.1 Current Compliance and 1T1R

As seen in Fig. 3.2, a compliance is used to control the current during the CF (dielectric breakdown). If the current is not limited during this step (or during set), the CF will be uncontrolled and too large which would negatively affect the reset and subsequent switching operations. The effective way to control the current during CF formation is with an imbedded (integrated) transistor in series with the memory cell (integrated 1T1R), which limits the current passing through the cell during the filament forming step (and set operations) thus preventing current from exceeding the desirable maximum level (current overshoot issue) [6, 7, 27–29]; however, conductivity overshoot may still occur [4] resulting in CF resistance being below the target value. By using an integrated transistor to control the filament formation (dielectric breakdown), the intrinsic properties of the RRAM devices can more clearly be revealed. The parasitic capacitance, which may also lead to current overshoot, can be limited, in general, by using small area drain, bottom and top electrodes, small and/or mesh-patterned probe pads, and thick isolation layers surrounding the RRAM cell. Using the special device structure shown in Fig. 3.4, which can allow for testing the same RRAM element in various different configurations of integrated 1T1R, or with external parametric analyzer, or even with an

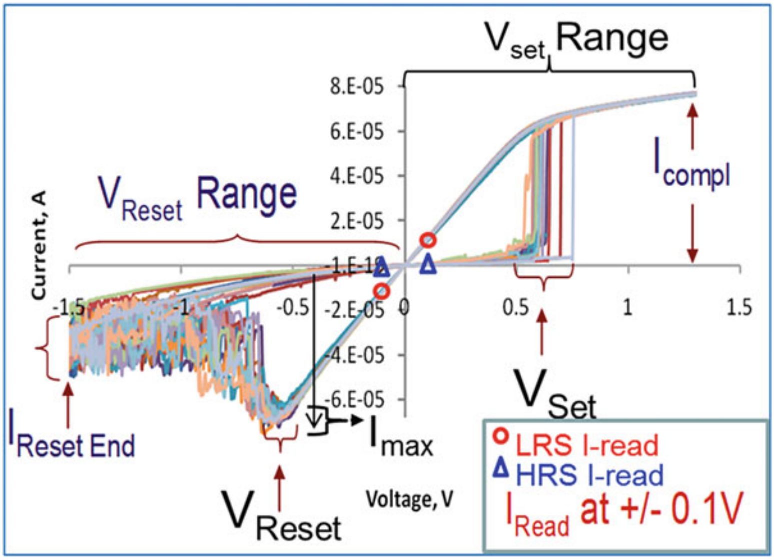

Fig. 3.3 A summary of the typical dc switching operation parameter terms for the bipolar-operated RRAM devices: Vset, V reset, Vrest range, Icomp, Imax, LRS I-Read, and HRS I-read. Vreset is defined by the onset of current reduction trend, Vreset range is the maximum voltage magnitude applied during the reset operation. Note that in DC operations, “Vreset range” is usually referred to as the reset voltage

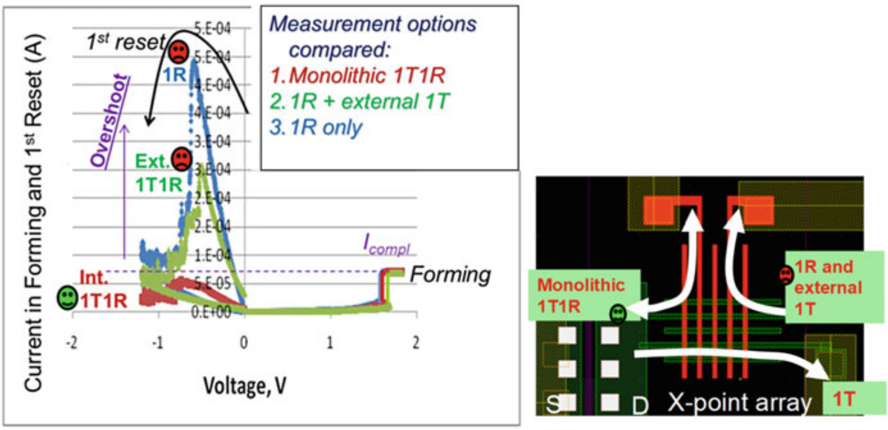

Fig. 3.4 Impact of parasitics on reset current. DC sweeps of forming and first reset on 100nmx100nm crossbar *TiN/Ti/HfOx/TiN* RRAM stack for the device configurations shown having compliance limited by integrated transistor (Int. 1T1R), externally connected transistor (Ext. 1T1R), or externally connected parametric analyzer (1R). Only the integrated transistor results in no current overshoot during first reset above the forming compliance value applied during CF formation

external transistor, to limit operation current, clearly demonstrates that only the integrated 1T1R configuration leads to elimination of the parasitic capacitance (as manifested in current overshoot of Imax above the compliance current [Icomp]) [7]. However, there are additional CF forming operation techniques, such as "multiforming" [30] and ultrafast pulse forming [4], which have demonstrated improved current control during the CF formation without the use of an integrated transistor. It should also be noted that there have likely been a large number of metal-oxide samples discarded upon poor RRAM switching, when they would have demonstrated stellar RRAM switching if only the current compliance during CF formation was properly controlled.

3.3 Material Properties Assisting Filament Formation

The properties of the initial CF created in the forming process determine the device switching characteristics. The successful forming process involves applying a sufficient bias across the metal-oxide dielectric resulting in the formation of a CF with ohmic- or near-ohmic-type conduction (as observed from the near linear I-V dependency after CF formation and increasing resistance with temperature). For the HfO_2 dielectric, these metallic characteristics indicate that the filament is represented by a Hf-rich/oxygen-deficient region in the dielectric. The filament formation is thus associated with the process of oxygen expulsion from this dielectric region, which is the focus of the following discussion. To start, an understanding of the stoichiometric and morphological properties of the HfO_2 dielectric film which are assisting formation of the conductive filament is required.

To assist the understanding toward connecting morphology and electrical response, conductive atomic force microscopy (C-AFM) can be an effective technique [31–33], allowing nanometer-resolved characterization of the electrical and topographical properties of oxide dielectrics [34]. When the C-AFM tip-sample system is biased, a tip scanning across the dielectric surface simultaneously generates data on both surface topography and current flow through the dielectric film. The C-AFM results show a good match between the maps of the leakage current and the grain boundaries (GBs) (Fig. 3.5) demonstrating that the current through the polycrystalline HfO_2 dielectric preferably flows along the grain boundaries, consistent with scanning tunneling microscopy (STM) measurements [35]. Dielectric breakdown, induced by the continuously applied voltage, preferentially occurs also at the GB sites. In addition, ab initio calculations were employed to identify the HfO_2 GB properties responsible for the current flow, and despite the GB stability, segregation of vacancies to the boundary is thermodynamically favorable [36].

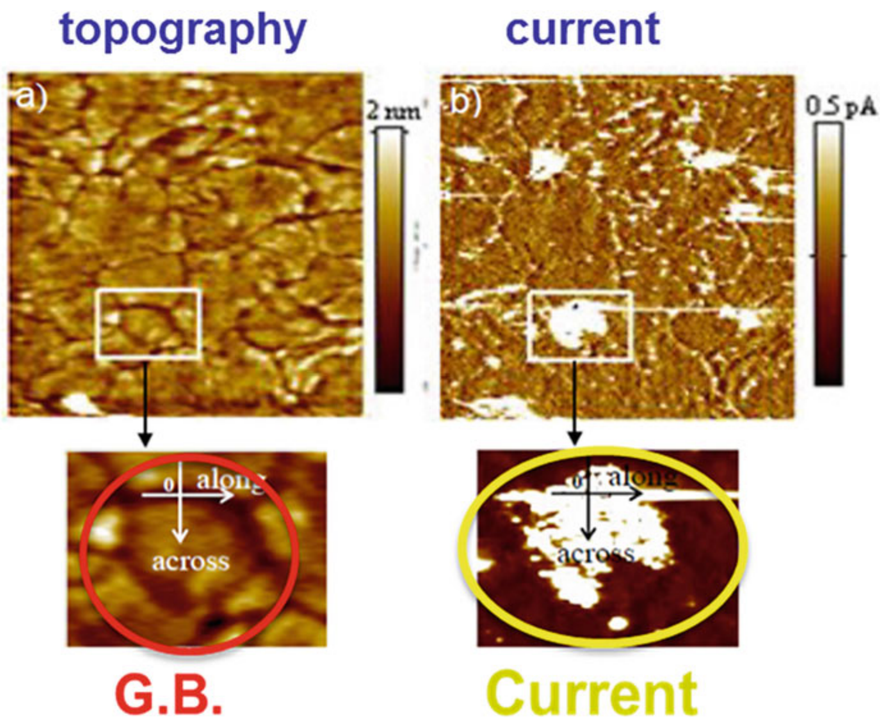

Fig. 3.5 Topographical (**a**) and current (**b**) images of a 5 nm HfO_2 film obtained with C-AFM. Multiple scans resulted in breakdown spots formed at the sites of the grain boundaries

3.3.1 The Role of Grain Boundaries in Electron Transport Through Hafnia

Electrical transport along the GBs in polycrystalline HfO_2 can be successfully described by considering the multi-phonon trap-assisted tunneling (TAT) process [26, 37] via vacancy sites. The leakage current through a TiN/HfO_2/TiN capacitor was simulated, considering both direct tunneling (DT) and TAT components of the gate current [36, 37], and successfully reproduced the current-voltage (I-V) temperature dependency in the polycrystalline HfO_2 [38]. The TAT current is the dominant component in dielectric films thicker than about 4 nm and is calculated by taking into account contributions from both the single-trap and multi-trap conductive paths by defects randomly placed along the GBs. The extracted values for the energy of the contributing traps and their relaxation energy match those calculated for the singly positively charged oxygen vacancy (V^+) at the GBs, indicating that the TAT current is associated with the (V^++e $\rightarrow V^0 \rightarrow V^+$+e) process [26]. The doubly positively charged oxygen vacancies (V^{2+}) exhibit the lowest diffusion activation energy (about 0.7 eV), and their segregation at the GBs leads to the formation of a conductive sub-band within the dielectric energy gap. By capturing injected

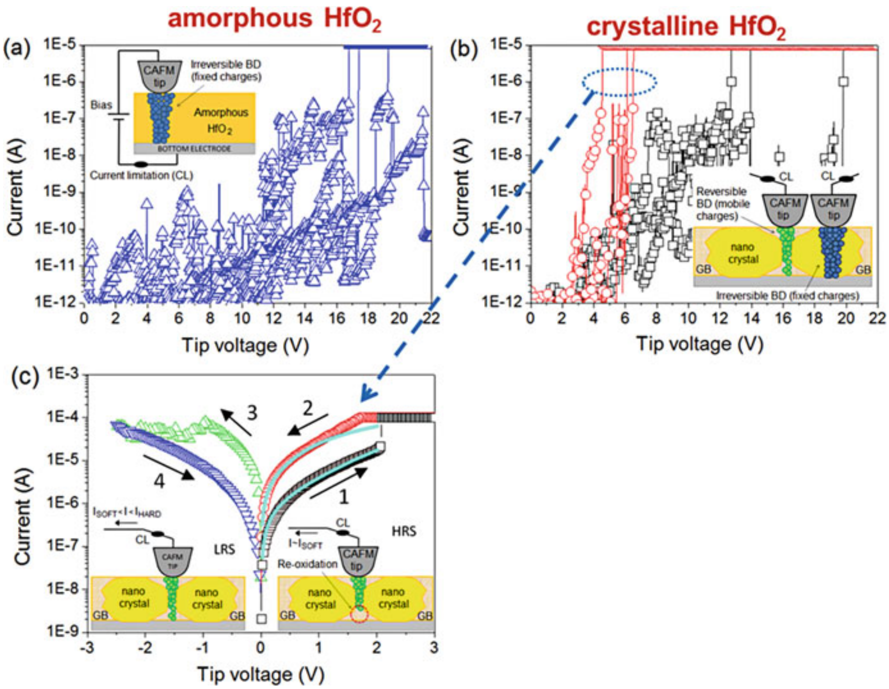

Fig. 3.6 Forming process at different random locations in (**a**) sample with non-annealed amorphous 3 nm HfO_2 and (**b**) annealed polycrystalline 3 nm HfO_2 where two different I-V patterns associated with low-voltage and high-voltage forming can be distinguished. (**c**) An example of switching observed at the low-voltage forming site in (**b**). The schematics indicate the probing location and measurement conditions

electrons when the forming voltage is applied, these vacancies are converted to a V^+ state, which is shown to support the TAT current by quickly capturing/releasing subsequently injected electrons [26]. The current flowing along the GBs thus facilitates a dielectric breakdown along one of the GB paths, which results in the preferential formation of the conductive filaments at the GB sites in polycrystalline metal oxides such as HfO_2.

To evaluate the effect of GBs on the filament formation, the forming/switching operations were performed in nanoscale by using the tip of a C-AFM probe as a top electrode of a MIM device (the biased TiN substrate constitutes the bottom electrode) (Fig. 3.6). The higher conductivity at the GBs in the polycrystalline film is accompanied by lower forming (breakdown) voltages. These lower forming voltage sites at GBs exhibit repeatable switching between high and low resistance states, while no resistive switching is observed when the forming is induced on the grain sites (as is also the case for an amorphous stoichiometric HfO_2), which exhibit significantly higher forming voltage values. These results indicate that resistive switching benefits from higher oxygen deficiency of the dielectric GBs making them easier to convert to a CF state at lower forming voltage [26].

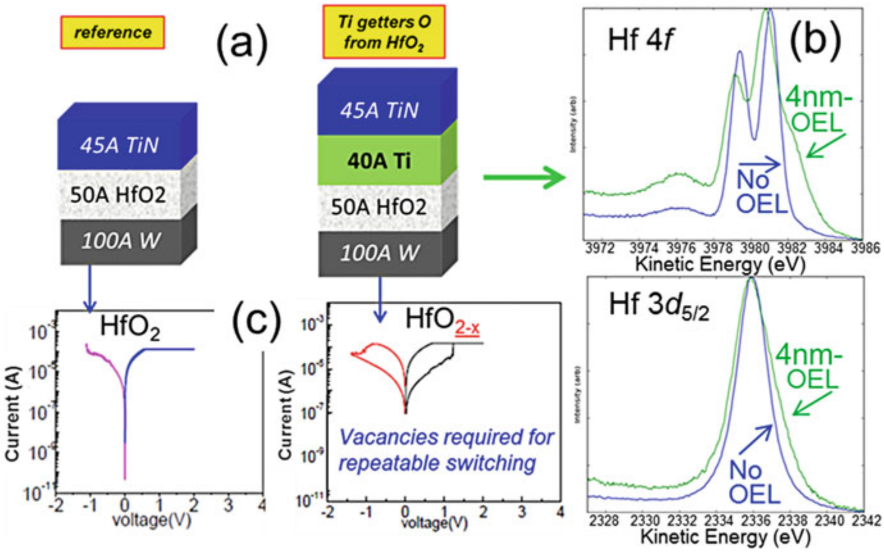

Fig. 3.7 Evidence of HfO_2 reduction by Ti-OEL; schematic of the stacks measured by the synchrotron XPS [at the National Synchrotron Light Source (NSLS) at (BNL)] (**a**), Hf 4f and $3d_{5/2}$ spectra (**b**), and switching characteristics of the corresponding devices (**c**)

It is also observed in MIM devices (such as TiN/HfO_2/TiN) that although perfect or near perfect stoichiometric HfO_2 does not readily demonstrate repeatable resistive switching after CF formation, the nonstoichiometric HfO_{2-x}, with "some" oxygen deficiency (density of vacancies), does exhibit resistive switching after forming. This is highlighted in Fig. 3.7 where an overlaying thin metal film can be used to scavenge or "getter" oxygen from stoichiometric HfO_2 dielectric leading to the oxygen deficiency as shown by the synchrotron XPS data. The sub-stoichiometric HfO_{2-x} film, with certain oxygen vacancies densities, demonstrates much better switching characteristics than the stoichiometric HfO_2 film. Thus, to create oxygen vacancies facilitating the CF formation, the gettering metal for extracting oxygens from the metal oxide is deposited over the metal-oxide film. Such gettering metal layers are usually early transitional metals (i.e., Ti, Zr, Hf, etc.) and often termed oxygen exchange layers (OEL) in RRAM stacks [26]. In addition, physical observations combined with electrical data have implicated HfO_2 GBs as susceptible toward reduction even when not favored from bulk thermodynamic considerations [39]. It is also noted that a typical thickness of a gettering metal in this application is 20A-100A and their gettering power grows with their thickness. Capping the gettering metal in a process that does not contain reactive nitrogen (which can reduce the metal gettering ability by nitriding it), like with in situ PVD tungsten after the gettering metal deposition, also increases the oxygen gettering ability of the OEL used.

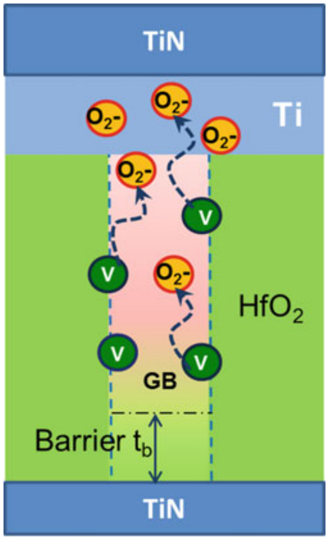

Fig. 3.8 Schematic illustrating oxygen ions diffusing out from the grain boundary to the overlaying OEL. The portion of the GB with higher oxygen vacancy density exhibits low resistance, while the GB portion further away from the OEL remains more stoichiometric, thus constituting an effective dielectric barrier

Using the above-described stack, the HfO_2 GB portion next to a gettering metal film is less resistive (due to higher concentration of the oxygen vacancies there), while a “more” stoichiometric or less “gettered” section of the GB away from gettering metal is a “better” more resistive dielectric due to less oxygen vacancies there (Fig. 3.8). Such a random process of oxygen gettering is expected to proceed differently in each GB due to variability of the GB characteristics (size, width, length, orientation, structure, etc.) resulting in a distribution of thicknesses of the “more” dielectric behaving barrier in the GB away from the gettering metal which would gate the current in the various GBs of each given device and, subsequently, a distribution of the magnitudes of leakage currents through these various GBs. The leakiest GB path is then accordingly expected to have the thinnest “dielectric” barrier since the electron transport through the barrier controls the overall current and the GB with the thinnest barrier should be the first one, out of all the other GB barriers with less conductive paths, to be broken by the applied voltage during CF forming. Thus, a single structural feature – the thinnest GB barrier in a given device – is responsible for both the lowest breakdown (CF forming) voltage magnitude and the highest leakage current (which is proportional to the total current through the device) and explains their correlation [26, 40]. Consequently, the primary beneficial effect of oxygen gettering is the creation of relatively highly conductive GB current paths (with higher oxygen deficiency toward the gettering metal), which can exhibit low forming voltages leading to a controlled CF formation that subsequently can enable the repeatable resistive switching phenomenon (i.e., continued rupture and forming of the CF).

3.4 Vacancy Asymmetry Engineering and the Metal-Oxide Filament-Based Switching Model

As described above, oxygen vacancies in sub-stoichiometric films may accumulate along the GBs in polycrystalline films [41], thus forming "preferred" conductive paths through the dielectric. The presence of the conductive paths along GBs assists with lowering the filament forming voltage allowing for more controlled and smaller conductive filament (CF) formation. Another important phenomenon of the CF is its nonuniform cross section resulting from the formation physics, giving it a cone-like shape (see Fig. 3.9a) [25, 26]. The cone-like CF shape provides certain advantages for switching operations, with bipolar reset described as oxidation of the narrower end of the CF tip to form a thin dielectric barrier near the positively biased reset-anode (which for bipolar operation is also near the forming cathode) leading to the high resistance state (HRS) (see Fig. 3.9b). The subsequent set operation (from HRS to the low resistance state [LRS]) is then the breakdown of this thin dielectric barrier thus reforming the CF. For a device to demonstrate well-behaved RRAM switching, the forming and switching biasing conditions should be selected according to the filament composition profile to achieve favorable distribution of the oxygen ions, which subsequently contribute to reoxidation of the filament tip during reset.

The *dc-iv* reset process in an HfOx-based RRAM device is illustrated in Fig. 3.10 [26, 41, 42]. Changes in the LRS during increasing reset bias are first observed by deviation off the linear I-V curve as could be expected for an ohmic LRS due to

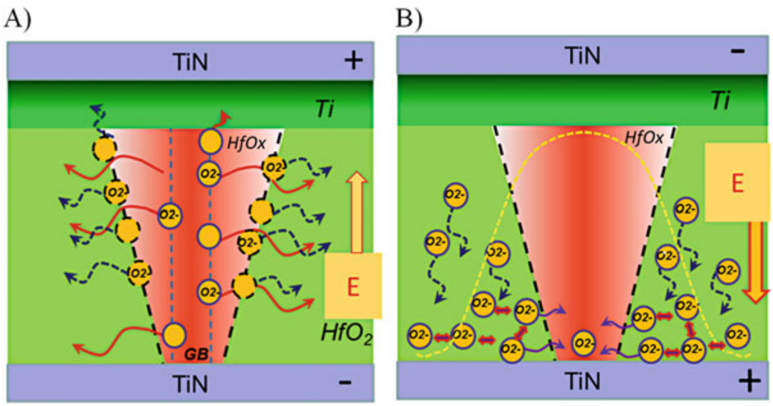

Fig. 3.9 (**a**) The filament formation process. The filament growth is initiated at GB (vertical broken lines) where the oxygen bonding is weaker than in the bulk oxide and proceeds via the process of the O(2-) ions release from the filament outer surface and moving with the electric field and also away from the filament due to concentration gradients. *These O-ions do not move too far as they are trapped by charges associated with grain boundaries and thus they are then readily available for CF oxidation in reset [26]. (**b**) For the reset process, the broken lines and broken bell-shape curve outline the filament and temperature radial profile, respectively. The nearby "available" O-ions diffuse following the electric field and density gradient and reoxidize a portion of the filament next to the positive anode electrode forming a thin HRS dielectric barrier

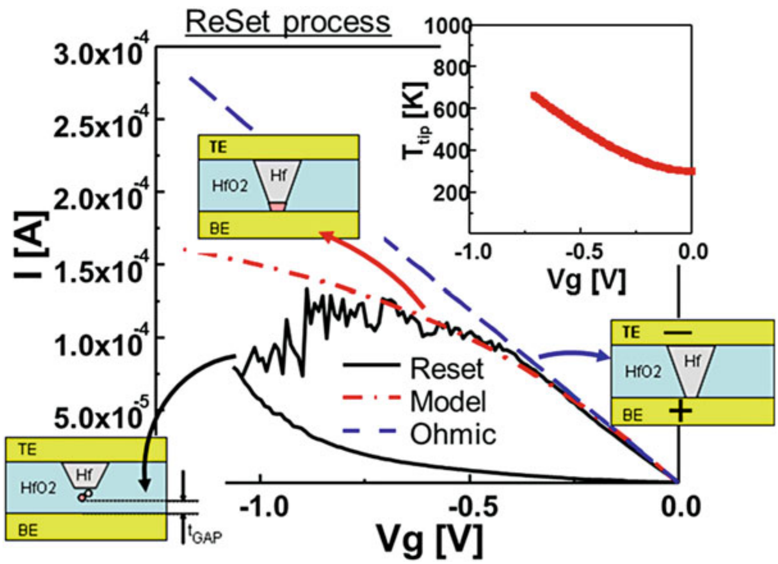

Fig. 3.10 RESET IV sweep with schematics of the physical process. Initially most voltage drops across the narrow conductive filament tip. The current deviates from ohmic initially due to a resistance increase from increasing filament temperature. Reset then occurs when the temperature is sufficient for oxidation and oxygen is available (i.e., diffusion from nearby "available" oxygen)

temperature increase of the CF (Fig. 3.10 inset shows the simulated temperature increase with increasing bias). Eventually upon continued bias and sufficient heat (due to the increasing current), oxidation of the CF tip begins, as observed by a "noisy" decrease in current with continued voltage increase. This fluctuating resistance magnitude reveals the random oxidation process near the CF tip where CF oxidation is competing with the process of generating new oxygen vacancy pairs in the dielectric barrier. This model of the reset process suggests that the vacancy concentration profile asymmetry in the RRAM dielectric stack is critically important for repeatable switching [26, 42–45] as discussed in this section below.

Robust switching in the metal-oxide-based RRAM devices is known to require some amount of oxygen deficiency of the dielectric film [26, 28, 45]; the reason for this requirement was discussed above primarily in regard to controlling CF formation at lower voltages. It has also been demonstrated that performance of RRAM devices is improved by utilizing asymmetric vacancy profiles of sub-stoichiometric metal-oxide films. These asymmetric vacancy profiles are typically formed using reactive PVD [11, 26, 28, 46], or with depositing a dielectric or metal and applying a subsequent posttreatment reduction or oxidation anneal [47–49], or by depositing reactive gettering-type metals (like described above and often termed oxygen exchange layers (OEL) in RRAM stacks) on to the metal oxide [9, 10, 14, 50, 51]. Many of the various metal-oxide-RRAM reports in the literature also consistently show bipolar operations with the reset bias applied with the positively biased reset-anode against the more oxygen-rich or more stoichiometric part of the engineered dielectric stacks [9, 10, 27, 45, 46, 48–50]. The reason the bipolar operation in metal-oxide filament-type RRAM consistently have this polarity preference characteristic is related to the metal-oxide asymmetric defect/vacancy profile, the CF asymmetry, and the switching mechanism as further described below. Having fewer defects/vacancies near the "reset-anode" allows for a more effective ability toward reoxidation of the narrow conductive filament tip in this area. For the case using an opposite bipolar operational polarity to switch the RRAM device (i.e., with the positively biased reset-anode against the more defects/vacancy-rich or less stoichiometric part of the engineered dielectric stack), too many nearby defects/vacancies and too few "available" oxygen atoms prevent effective oxidation of the CF tip [26, 45].

3.5 Switching in Asymmetric Vacancy Engineered RRAM

To investigate the preferred bipolar operational biasing and vacancy asymmetry relationship, various sub-stoichiometric HfO_x films with asymmetric oxygen vacancy profiles were formed [26, 45]. An OEL of titanium was deposited on top HfO_2 and subsequently capped in situ with TiN creating an asymmetric defect/vacancy profile with more defects/vacancies near the OEL side of the stack due to preferential gettering of oxygen from the HfO_2 grains and grain boundaries nearer the OEL metal. Figure 3.11a, b presents the switching behavior for the two possible bipolar biasing schemes: (1) Fig. 3.11a, with CF formed in a bipolar biasing scheme that has the positively biased reset-anode opposite the OEL (or more defects/vacancies) side, and (2) Fig. 3.11b, with CF formed in a bipolar biasing scheme that has the positively biased reset-anode at the OEL (or more defects/vacancies)

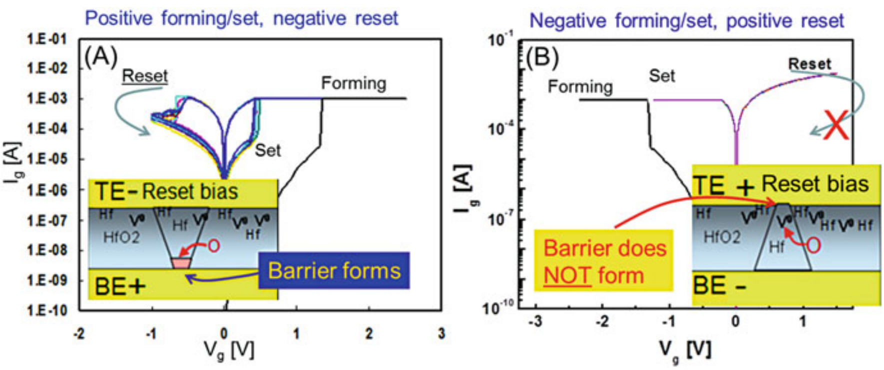

Fig. 3.11 Results for applying the two possible bipolar operational biasing schemes. When operational biasing has the reset-anode near the more oxygen-rich or stoichiometric side, a dielectric barrier forms successfully achieving reset (**a**). However, having oxygen-poor or higher vacancy/defect concentration near the reset-anode prevents a dielectric barrier from forming, and no reset is observed (**b**)

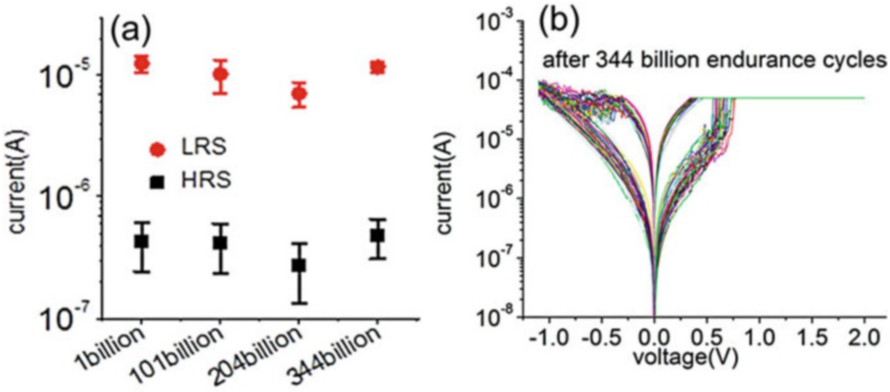

Fig. 3.12 Endurance results for similar RRAM stack as used for Figs. 3.10 and 3.11 (TiN[HfO_x\Ti-OEL]TiN). Averaged LRS and HRS reads from 50 dc cycle checks at 1, 101, 204, and 344 billion ac endurance cycles are charted for I_{comp} = 60uA, pulse set voltage = 1.5 V, pulse reset voltage = −1.3 V, pulse rise/fall time = 15 ns, and pulse top = 5 ns (**a**). The 50 cycle dc check after 344 billion cycles is shown in 12(**b**)

side. Only operating in the mode with a bias polarity having the positively biased reset-anode against the more stoichiometric part of the film (away from the OEL) results in well-behaved RRAM switching with well over 100 billion ac cycles demonstrated on device dimensions down to 50nmx50nm (see Fig. 3.12). Operating the device opposite this preferred polarity often results in the failure to reset toward higher resistance states (Fig. 3.11b) due to difficulty in reoxidation of the CF tip because of the high defect/vacancy density near the reset-anode in this "non-preferred" bipolar biasing case.

This oxygen vacancy asymmetry and preferred bipolar biasing is demonstrated further with a RRAM device using a decidedly asymmetric HfOx film formed by oxidizing the top of an exceedingly sub-stoichiometric HfOx film. The backside-SIMS profile of the oxygen concentration for the further oxidized HfOx film is shown in Fig. 3.13 indicating the film has high defect/vacancy concentration near the bottom TiN electrode and near stoichiometric HfO_2 near the top TiN electrode.

Note the defect profile for this stack in Figs. 3.13 and 3.14 (defects/vacancies near bottom electrode) is opposite to that of the engineered HfOx\OEL described above (with defects\vacancies near the OEL and top electrode). The ramifications of this opposite defect profile in the "top"-oxidized HfOx with respect to the HfOx\OEL ("top"-"reduced") stack, is that now the preferred bipolar operational biasing (i.e., having the positive reset-anode against the more stoichiometric side of the film), occurs with negative forming/set, and positive reset, voltages applied to top electrode. In addition, when performing the preferred operational biasing for this highly asymmetric "top"-oxidized HfOx, not only are over 10^8 *ac-pulsed* endurance cycling and 100% *dc-iv* switching yield observed (Fig. 3.14), but operating with opposite this "preferred" bipolar biasing, with reset-anode against the defective/vacancy-rich side of the HfOx film, the resulting switching yield was zero%! Thus, RRAM

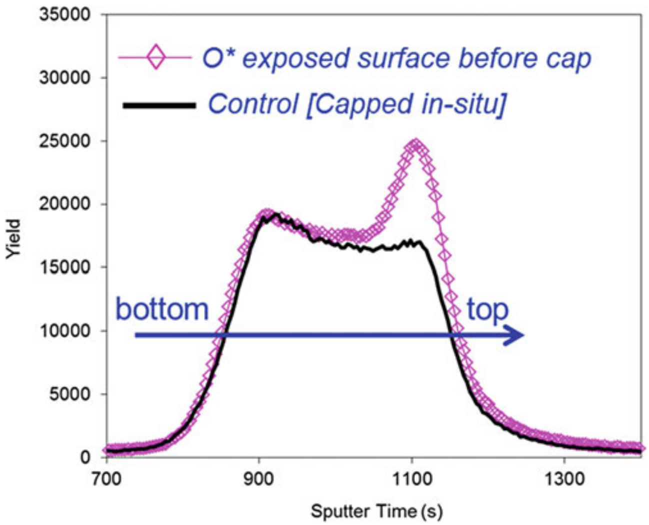

Fig. 3.13 Backside SIMS showing the oxygen profiles for highly sub-stoichiometric HfO_x with and without exposing the top surface to oxygen plasma before capping with Pt (Pt used only for SIMS analysis)

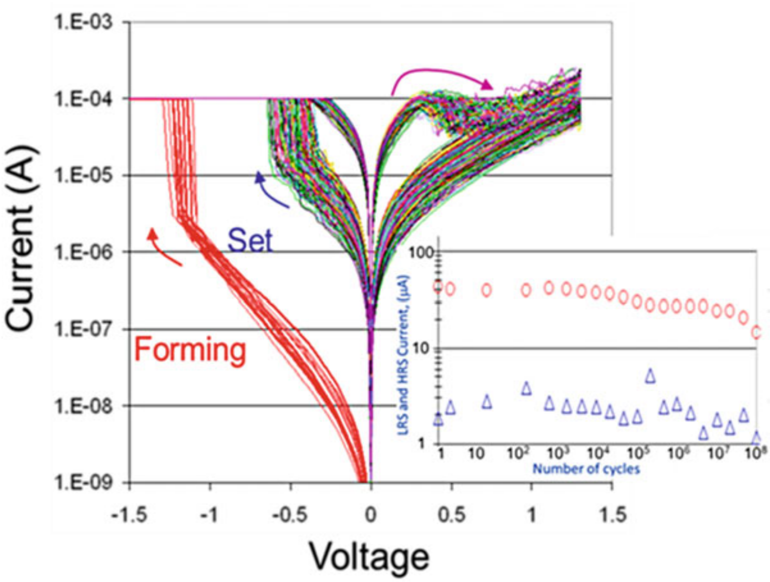

Fig. 3.14 Over 10^8 *ac* endurance cycling (inset) and 100% *dc* switching yield observed when performing the preferred operational biasing for a highly asymmetric oxidized HfO_x film (with TiN electrodes and external compliance control). The dc switching shown is for 350 reset/set sweeps (for all 35 die tested x 10 cycles each)

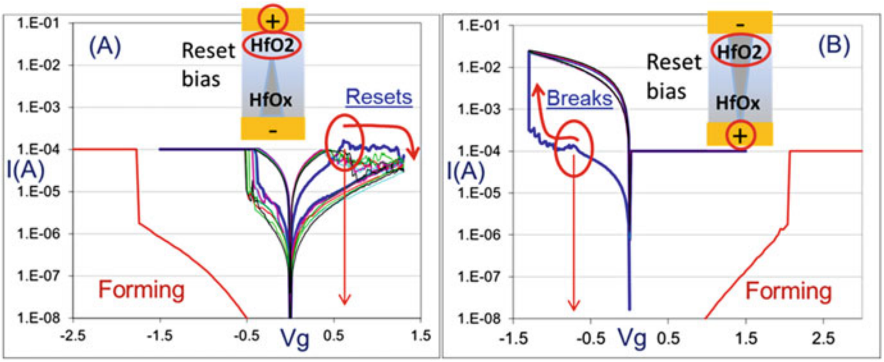

Fig. 3.15 Typical dc forming and cycling with asymmetric HfO_x for (**a**) preferred bipolar biasing (reset-anode against "good" more stoichiometric HfO_2) vs (**b**) non-preferred biasing (reset-anode against "bad" vacancy-rich HfO_x)

operation for this highly asymmetric sub-stoichiometric metal-oxide film dramatically illustrates the results of the "preferred" operational bipolar biasing relationship to the metal-oxide vacancy/defect profile and also how the oxygen vacancy profile supports the described reset mechanism of CF-tip oxidation. Having the positive reset-anode against the defective/vacancy-rich part of the HfOx stack allows for too many available conduction paths and insufficient oxidation of the CF for a HRS tunnel barrier to form. Furthermore, comparing *dc-iv* reset sweeps for the two bipolar biasing cases of reset-anode against "good" (more stoichiometric) or "bad" (more oxygen vacancies) part of the dielectric, it is observed that even for the "non-preferred" bias polarity with positively biased reset-anode against the "bad" vacancy-rich part of the dielectric, a similar deviation in current near reaching the forming compliance current value can be seen (Fig. 3.15). However, for this non-preferred biasing of the asymmetric vacancy stack, the current does not continue a downward trend to HRS (CF-tip oxidation) but begins to increase and the device "breaks" (with runaway current due higher rate of vacancy formation vs oxidation leading to a "breakdown," and with no compliance in reset, a resulting uncontrolled large CF is formed). Thus, the reset case with too many defects near the reset-anode suppresses the ability for CF-tip oxidation and denies a competition between filament oxidation and breakdown leading to only further breakdown (or CF size increase) and the resulting reset failure. Engineering the oxygen vacancy profile to support the described forming, reset, and set mechanisms is required for RRAM performance optimization.

Similar predisposition as the HfOx-based system for a preferred bipolar operation biasing and associated engineered oxygen vacancy asymmetry has been demonstrated for TaOx-based RRAM. Figure 3.16a shows *dc-iv* behavior for a TaOx-based stack consisting of a deposited thin layer of tantalum metal followed in situ by a deposited thicker and highly sub-stoichiometric TaOx layer, followed by a deposited thin stoichiometric Ta_2O_5 layer on top. The "preferred" bipolar operation of this stack is as expected with the positive reset-anode against the "good," or more

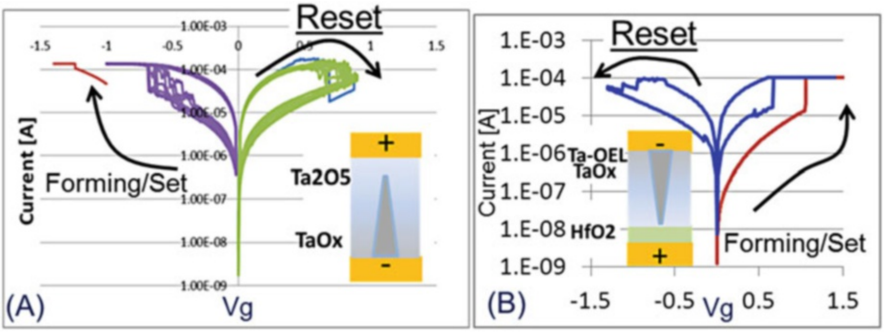

Fig. 3.16 The dc forming/cycling for (**a**) asymmetric [bottom\Ta\TaO$_x$\Ta$_2$O$_5$\top] with "preferred" negative forming/set and positive reset applied to top electrode and (**b**) asymmetric [bottom\HfO$_2$\TaO$_x$\Ta-OEL\top] with positive forming/set and negative reset applied to top electrode

stoichiometric, "top" Ta_2O_5 side of the stack (so the forming was implemented with the negative cathode against the top electrode). Furthermore, a stack having near 1 nm stoichiometric HfO_2 deposited and then capped with a thick sub-stoichiometric TaOx layer, and then a thin tantalum metal-OEL layer (Fig. 3.16b) demonstrates similar bipolar switching as the RRAM stack with a single thicker HfO_2 with "top" deposited OEL of Figs. 3.10, 3.11, and 3.12 where preferred positive reset-anode is opposite the vacancy-rich OEL\top electrode (and against the more stoichiometric side of the stack). The similar reset for these devices with different metal-oxide RRAM material but similar vacancy profile stacks also supports the reset mechanism of oxidizing the CF tip near the reset-anode toward a thin (~1 nm) dielectric barrier [41]. The similar behaving films in Figs. 3.11a and 3.16b, which both have HfOx at the positive reset-anode, are both forming a similar ~1 nm HfO_2 tunnel barrier in reset with similar I-V characteristics [26, 45].

3.6 Real-Time Monitoring of RRAM Switching Highlights Operating Mechanism

These above-described relevant mechanisms for the bipolar metal-oxide RRAM operation can be highlighted by capturing the changes in the device conductivity in real time while applying "set or reset" type *ac* bias signals to the RRAM stack [6, 7, 26]. For instance, in general, a dielectric barrier limiting the current in a HRS should break similarly irrespective of polarity of the applied bias across the dielectric barrier; however, there is a clear difference to the HRS conductivity response when pulsed voltages of the same pulse width and height, but opposite polarities, are applied across the dielectric formed at the CF tip [26, 45]. For a "preferred" set-type bias (with a positive anode biasing near the defect\vacancy-rich side) applied to the device in the HRS, a clear abrupt change to a lower resistance state occurs

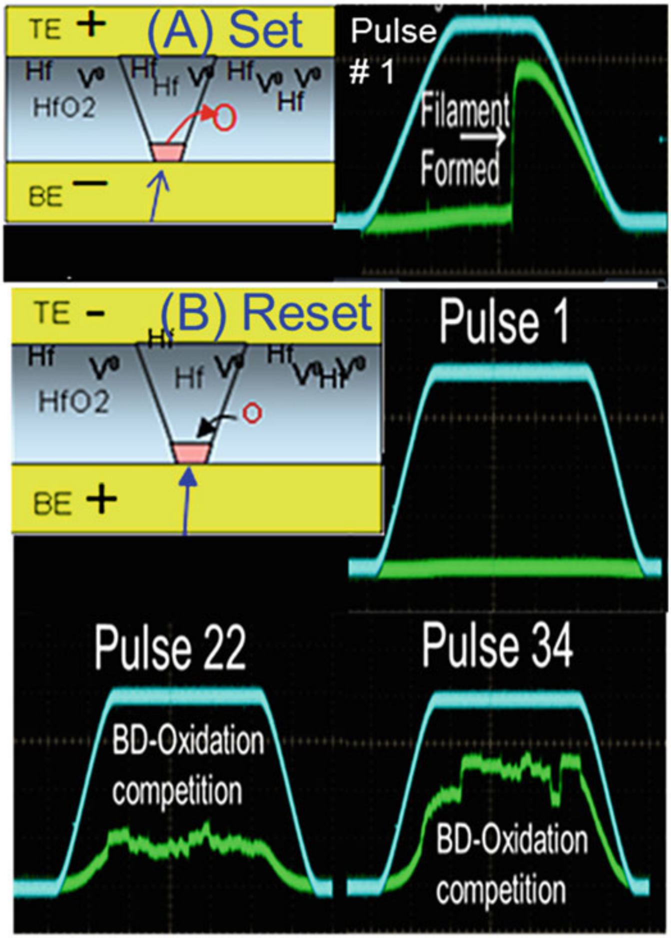

Fig. 3.17 (**a**) Set pulse (100us and 0.8 V) applied to HRS state of the RRAM stack (similar RRAM stack as used for of Figs. 3.10, 3.11, and 3.12). Immediate and singular resistance change to LRS is observed indicating reformation of the CF for the set bias. (**b**) Reset pulse (100us and −0.8 V) applied to HRS state of the RRAM stack where many pulses are required for resistance change, and then the fluctuating resistance change is reflecting the competition between HRS breakdown and reoxidation

(Fig. 3.17a): the dielectric barrier abruptly breaks with the released negatively charged oxygen ions being pushed away from this dielectric barrier (that was formed at the CF tip in reset) and toward the positive anode by the electric field. Due to lack of “available” oxygen near the “breakdown” in the dielectric barrier and a higher oxygen vacancy concentration (from forming bias choice and the asymmetric oxygen vacancy engineering of the fabricated metal-oxide stack) near the positively biased anode for this “preferred” set-type biasing, the dielectric barrier breakdown occurs readily and immediately (and no dielectric barrier is formed at the opposite side where the positive voltage has been applied due to high vacancy concentration there). For the preferred reset-type bias (positive anode bias opposite the defect \vacancy-rich side and near the more stoichiometric side) applied to the device in the HRS, many continuous pulses are required before any resistance change occurs (Fig. 3.17b). In addition, when the resistance does change, the signal is very noisy with no immediate sharp current increase. According the above-discussed switching model, an apparent randomness of fluctuations of the HRS resistance observed under the continued "preferred reset-type" pulses results from a competition between the Hf-O bond breakage in the dielectric barrier and the near immediate reoxidation of the Hf atoms caused by a continued presence of the "available" oxygen ions in their vicinity: the applied field pushes these ions toward the positively biased electrode, thus preventing their fast out-diffusion from the filament region and providing ample available oxygen for filament tip oxidation in this area.

Figure 3.18 shows oscilloscope-captured switching data for a pulsed width of 20 ns for set, and 70 ns for reset, in fully integrated 1T1R HfO_x-OEL RRAM

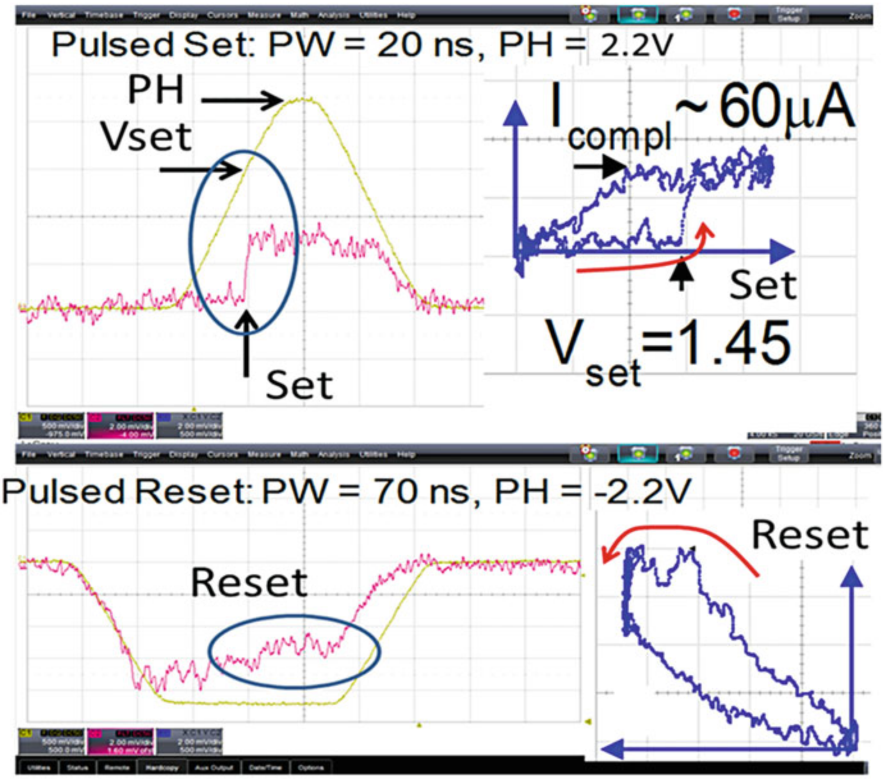

Fig. 3.18 Pulsed switching in fully integrated 1T1R HfO_x\Ti-OEL RRAM for 100 nm × 100 nm cross-bar devices. Yellow, pulse voltage applied to the top electrode; red, voltage at the bottom electrode as read through the oscilloscope: inserts (blue), pulsed I-V traces corresponding to the respective pulsed set and reset operations

100 nm × 100 nm cross-bar devices (similar RRAM stack as used in Figs. 3.10, 3.11, and 3.12) [45]. The switching for both set and reset between HRS and LRS in the ns time frame is clearly observed. Additionally, for the "preferred" operational biasing applied, even at this ns time scale, the immediate and singular resistance change is observed for set operation (a dielectric breakdown), and the more noisy increase/decrease fluctuations in resistance are observed for reset operation (due to the reset bias additionally having an effective competition between oxidation of the CF tip to form a dielectric barrier and a breakdown of this developing dielectric barrier).

3.7 Conclusion

Optimizing metal-oxide RRAM switching requires an understanding of how engineering of asymmetric oxygen vacancy distribution and filament geometry assist the operation preferences of the bipolar switching schemes. In general, the vacancy profile will have one side of the metal-oxide dielectric more vacancy-rich/oxygen-poor with the opposite side more vacancy-poor/oxygen-rich, and the preferred bipolar biasing of this cell is such that reset more effectively occurs when the positive anode is next to the vacancy-poor/oxygen-rich side. The goal is creating this oxygen vacancy profile to be advantageous for controlled CF formation (and reformation in set operations), along with dielectric barrier formation (from CF-tip oxidation) during reset, under the conditions benefiting high-speed/low-power switching.

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CHAPTER 1

Introduction to RRAM Technology

1.1 OVERVIEW OF EMERGING MEMORY TECHNOLOGIES

The functionality and performance of today's computing system are increasingly dependent on the characteristics of the memory sub-system. The memory sub-system has a well-known hierarchy: From top layers to bottom layers, SRAM, DRAM, and FLASH are the mainstream memory technologies serving as cache, main memory, and solid-state-drive (SSD), respectively. Moving up the hierarchy, the memory write/read latency decreases. Moving down the hierarchy, the memory capacity increases. All these mainstream memory technologies are based on the charge storage mechanism: SRAM stores the charge at the storage nodes of the cross-coupled inverters, DRAM stores the charge at the cell capacitor, and FLASH stores the charge at the floating gate of the transistor. All these charge-based memories are facing challenges to be scaled down to 10 nm node or beyond, due to the easy loss of the stored charge at nanoscale, resulting in the degradation of the performance, reliability, and noise margin, etc. In this context, emerging memory technologies that are non-charge based are actively under research and development in the industry, with the hope to revolutionize the memory hierarchy [1].

The ideal characteristics for a memory device include fast write/read speed (<ns), low operation voltage (<1 V), low energy consumption (~fJ/bit for write/read), long data retention time (>10 years), long write/read cycling endurance (>10^{17} cycles), and excellent scalability (<10 nm). Nevertheless, it is almost impossible to satisfy these ideal characteristics in a single "universal" memory device. Several emerging non-volatile memory (NVM) technologies have been pursued toward to achieving part of these ideal characteristics. The emerging NVM candidates are spin-transfer-torque magnetoresistive random access memory (STT-MRAM) [2], phase change random access memory (PCRAM) [3], and resistive random access memory (RRAM) [4]. These emerging NVM technologies share some common features: they are non-volatile two-terminal devices, and they differentiate their states by the switching between a high resistance state (HRS, or off-state) and a low resistance state (LRS, or on-state). The transition between the two states can be triggered by electrical inputs. However, the detailed switching physics is quite different for different memories: STT-MRAM relies on difference in resistance between the parallel configuration (corresponding to LRS) and the anti-parallel configuration (corresponding to HRS) of two ferromagnetic layers separated by a thin tunneling insulator layer; PCRAM relies on chalcogenide material to switch between the crystalline phase (corresponding to LRS) and the amorphous phase (corresponding to HRS); and RRAM relies on the formation (corresponding to LRS) and the rupture (corresponding to HRS) of conductive filaments in the insulator between two electrodes. Due to the different underlying physics, the device characteristics are also different among different emerging NVM technologies. Table 1.1 compares the typical device characteristics between the emerging memory technologies and the mainstream memory technologies. It should be pointed out that different emerging NVM devices may have different application spaces due to their unique characteristics. As is seen from Table 1.1, compared to SRAM, STT-MRAM has an advantage of a smaller cell area, while STT-MRAM has maintained low programming voltage, fast write/read speed, and long endurance, thus STT-MRAM is attractive for embedded memories on chip, e.g., the SRAM replacement in the last-level cache [5]. Compared to FLASH, RRAM is attractive due to its lower programming voltage and faster write/read speed, thus the primary target of RRAM is to replace the NOR FLASH for code storage and more ambitiously to replace NAND FLASH as data storage [6]. Besides replacing the existing technologies, the emerging NVM technologies hold the potential to revolutionize today's memory hierarchy by adding more levels in the hierarchy, e.g., creating a storage class memory level between the main memory and the storage memory [7]. In addition, a hybrid system with emerging memories and mainstream memories is also attractive, e.g., using RRAM as the cache for NAND FLASH [8].

Table 1.1: Device characteristics of mainstream and emerging memory technologies

	Mainstream Memories				Emerging Memories		
	SRAM	DRAM	FLASH		STT-MRAM	PCRAM	RRAM
			NOR	NAND			
Cell Area	$>100F^2$	$6F^2$	$10F^2$	$<4F^2$ (3D)	$6\sim20F^2$	$4\sim20F^2$	$<4F^2$ if 3D
Multi-bit	1	1	2	3	1	2	2
Voltage	<1V	<1V	>10V	>10V	<2V	<3V	<3V
Read Time	~1ns	~10ns	~50ns	~10μs	<10ns	<10ns	<10ns
Write Time	~1ns	~10ns	10μs-1ms	100μs-1ms	<5ns	~50ns	<10ns
Retention	N/A	~64ms	>10y	>10y	>10y	>10y	>10y
Endurance	>1E16	>1E16	>1E5	>1E4	>1E15	>1E9	>1E6~1E12
Write Energy (J/bit)	~fJ	~10fJ	100pJ	~10fJ	~0.1pJ	~10pJ	~0.1 pJ

F: feature size of the lithography, and the energy estimation is on the cell-level (not the array-level)

1.2 RRAM BASICS

The resistive switching phenomenon where the resistance of insulators such as metal oxides changes when a large voltage is applied was firstly reported in the 1960s [9]. The recent revival on the resistive switching can be traced back to the discovery of hysteresis I-V characteristics in perovskite oxides such as $Pr_{0.7}Ca_{0.3}MnO_3$ [10], $SrZrO_3$ [11], $SrTiO_3$ [12], etc., in the late 1990s and the early 2000s. Since Samsung demonstrated NiO RRAM array integrated with the 180 nm silicon CMOS technology in 2004 [13], research activities have been blooming with demonstrations of resistive switching in various binary oxides[1] such as NiO [14], TiOx [15], CuO_x [16], ZrO_x [17], ZnO_x [18], HfO_x [19], TaO_x [20], AlO_x [21], etc., because of the simplicity of the materials and good compatibility with silicon CMOS fabrication process. Later in 2008, HP Labs made the connection of the resistive devices to the theoretical concept of memristor [22].[2]

Generally speaking, there are two types of RRAM. The first type is based on the conductive filaments consisting of oxygen vacancies, which is typically referred to as oxide-based RRAM; the second type is based on the conductive filaments consisting of metal atoms, which is also called conductive-bridge RAM (CBRAM). CBRAM relies on the fast-diffusive Ag or Cu ions migration into the oxide (or chalcogenide) to form a conductive bridge. Despite different underlying switching physics, these two types share a lot of common device characteristics and the array architecture design considerations are very similar. In this lecture, we focus on the first type: oxygen vacancy based oxide RRAM.[3] In literature, there are a few comprehensive reviews on the oxide RRAM [23, 24, 25, 4]. For the CBRAM, refer to the review [26].

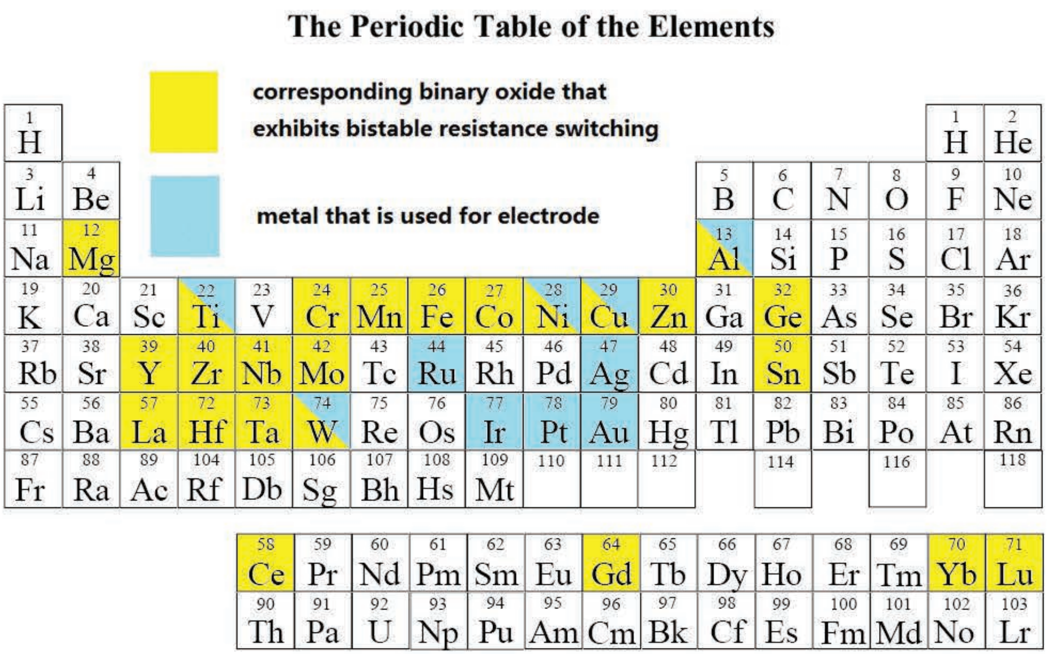
So far, tens of binary oxides have been found to exhibit resistive switching behavior. Most of them are transition metal oxides, and some are lanthanide series metal oxides. The materials for the resistive switching oxide layer and the electrodes reported in literature are summarized in Table 1.2. Besides metals, conductive nitrides, e.g., TiN, TaN, are also commonly used as electrode materials.

[1] These binary oxides that show resistive switching are often non-stoichiometric, thus subscript "x" is used for the oxygen composition in this book.

[2] To unify the terminology and emphasize the technology development, RRAM is used instead of memristor in this book.

[3] If not specifically noted, the term "RRAM" refers to the binary oxide memory involving oxygen vacancies in this book.

Table 1.2: Summary of the materials that have been used for binary oxide RRAM reported in literature . Metals of the corresponding binary oxides used for the resistive switching layer are colored in yellow. Metals used for the electrodes are colored in blue. Used with permission from [4].



Before starting the discussion in this lecture, we first introduce some basic concepts and terminologies about RRAM. Figure 1.1 (a) shows the typical metal-insulator-metal device structure of RRAM: a thin oxide layer sandwiched between two electrodes. The switching event from HRS to LRS is called the "set" process. Conversely, the switching event from LRS to HRS is called the "reset" process. Usually for the fresh samples, the initial resistance is very high and a large voltage is needed for the first cycle to trigger the switching behaviors for the subsequent cycles. This is called the "forming" process. The switching modes of RRAM can be broadly classified into two switching modes: unipolar and bipolar. Figure 1.1 (b) and (c) show a sketch of the I-V characteristics for the two switching modes. Unipolar switching means the switching direction depends on the amplitude of the applied voltage but not on the polarity of the applied voltage. Thus set/reset can occur at the same polarity. If the unipolar switching can symmetrically occur at both positive and negative voltages, it is also referred to as a nonpolar switching mode. Bipolar switching means the switching direction depends on the polarity of the applied voltage. Thus set can only occur at one polarity and reset can only occur at the reversed polarity. For either switching mode, to avoid a permanent dielectric breakdown in the forming/set process, it is recommended to enforce a compliance current, which is usually provided by the semiconductor parameter analyzer during off-chip testing, or more practically, by a cell selection device (transistor, diode, or a series resistor) on-chip. To read the data from the cell, a small read voltage that does not affect the memory state is applied to detect whether the cell is in HRS or LRS.

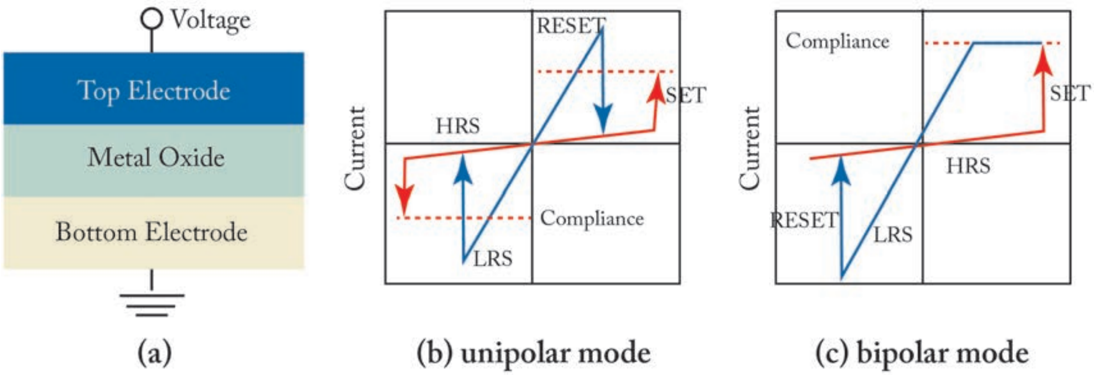


Figure 1.1: (a) Schematic of metal-insulator-metal structure for oxide RRAM, and schematic of RRAM's I-V curves, showing two modes of operation: (b) unipolar and (c) bipolar. Adapted from [4].

It is observed that the electrode materials have a significant impact on the switching modes of the oxide RRAM. Even with the same oxide material but with different electrode materials, the switching modes can be different. Therefore, it is inferred that the switching mode is not an intrinsic property of the oxide itself but a property of both the oxide materials and the electrode/oxide interfaces. In most cases, the unipolar mode is obtained with a noble metal such as Pt as both top and bottom electrodes. With one of the electrodes replaced by oxidizable materials such as Ti or TiN, the bipolar mode is obtained. For the bipolar switching, it was suggested that the reversed field is required for a successful reset because an interfacial oxygen barrier (e.g., TiON) exists [27]. If both electrodes are oxidizable, there should be some asymmetry in the oxygen gettering capability for the bipolar switching. A typical structure is TiN/metal/oxide/TiN,[4] e.g., TiN/Ti/HfO_x/TiN [19], TiN/Hf/HfO_x/TiN [28], where the metal capping layer functions as an oxygen gettering layer. Generally, the unipolar switching needs a higher reset current than the bipolar switching, and shows a larger variability as well. Therefore, today RRAM research and development focus more on the bipolar switching, and we will focus our discussions on the bipolar mode in this book.[5]

To further understand how the RRAM device works as NVM, the device characteristics of HfO_x-based RRAM [19, 29] from Industrial Technology Research Institute (ITRI) at Taiwan are presented here as an example. Figure 1.2 (a) shows the transmission electron microscopy (TEM) image of the concave structure device of TiN/Ti/HfO_x/TiN stack with 30 nm cell size. Figure 1.2

4 If not specifically noted, the stack sequence is from the top electrode to the bottom electrode in this book.
5 If not specifically noted, the switching mode is bipolar switching in this book.

(b) shows the typical I-V curve of this RRAM cell. A 200 μA set compliance current is enforced, and the device exhibits a bipolar switching. Figure 1.2 (c) shows the programming cycling endurance characteristics. The set/reset programming condition is +1.5 V/-1.4 V pulse with 500 μs width, and after 10^6 switching cycles the resistance on/off ratio is still larger than 100. Figure 1.2 (d) shows the data retention testing results. The device is baked at 150°C, and i a 10-year lifetime is expected using a simple linear extrapolation.

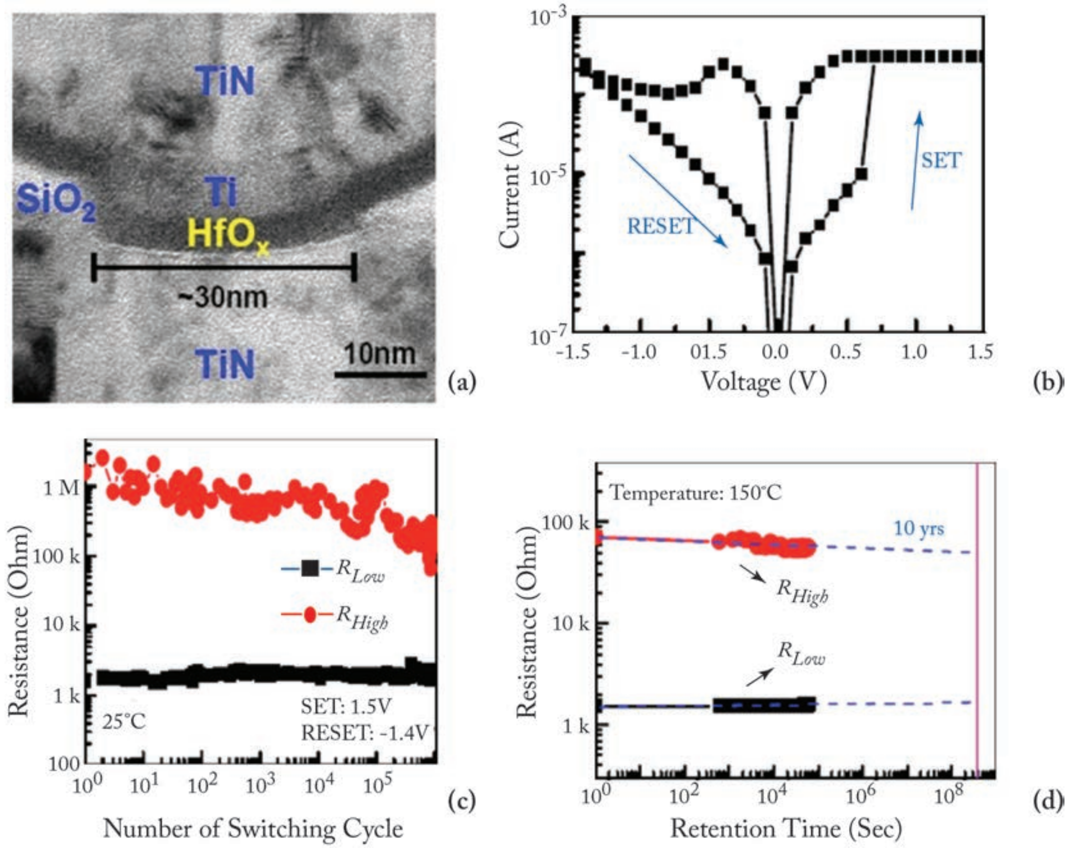


Figure 1.2: (a) Transmission electron microscopy (TEM) image of the concave structure device of ITRI's TiN/Ti/HfOx/TiN stack; (b) typical I-V curve of the device with the cell size of 30 nm; (c) programming endurance cycling by 500 μs pulse: successive 10^6 switching cycles is achieved; (d) retention testing at 150 °C: a 10-year lifetime is expected. Adapted from [19, 29].

1.3 RECENT RESEARCH AND DEVELOPMENT OF RRAM TECHNOLOGY

The development of oxide RRAM has progressed rapidly in the past decade. In particular, binary oxides that use materials that are compatible to the silicon CMOS fabrication process have seen intense research and development in the industry. The early RRAM devices in the mid-2000s had large device area (>>μm^2), large programming current (~mA), long programming time (>μs), low endurance (<10^3 cycles), and required a large forming voltage (~10 V). Today, many of these deficiencies have been overcome. Device sizes down to 10 nm or below have been demonstrated [28, 30], programming current is now in the order of tens of μA or a few μA, programming speed is in the order of tens of ns or a few ns, programming endurance cycles are typically larger than 10^6 with a record up to 10^{12} [31], retention time is >3,000 h at 150°C and extrapolated to be more than 10 years at 85°C [20], and the forming process can be eliminated by shrinking the oxide thickness [32] or other oxide stack engineering strategies. Most of these good characteristics were reported in HfO_x or TaO_x systems. Demonstrations of 2-bit and 3-bit multi-level operation have also been made [33, 34]. Chip-level RRAM array macro from 4 Mb to 32 Gb capacity with peripheral circuitry have been demonstrated by industry as well [35, 36, 37], showing that RRAM is a viable NVM technology for practical applications.

This book is organized as follows. Chapter 2 will discuss the RRAM device fabrication techniques and methods to eliminate the forming process, and will show its scalability down to sub-10 nm regime. Then the device performances such as programming speed, variability control, and multi-level operation will be presented, and finally the reliability issues such as cycling endurance and data retention will be discussed. Chapter 3 will discuss the RRAM physical mechanism, the materials characterization techniques to observe the conductive filaments, and the electrical characterization techniques to study the electronic conduction processes. It will also present the numerical device modeling techniques for simulating the evolution of the conductive filaments as well as the compact device modeling techniques for circuit-level design. Chapter 4 will discuss the two common RRAM array architectures for large-scale integration: one-transistor-one-resistor (1T1R) and cross-point architecture with selector. The write/read schemes are presented and the peripheral circuitry design considerations are discussed. Finally, a 3D integration approach is introduced for building ultra-high density RRAM array. Chapter 5 is a brief summary and will give an outlook for RRAM's potential novel applications beyond the NVM applications.